Correct the claim.
Keep the valid mathematics.

A worked contribution for Oleksiy Babanskyy's PAPP

Prepared for Alexandre Couret with AI assistance 24 October 2026

Purpose. Provide a small, reusable mathematics example for the rules PAPP already states: exact scope, surviving results, evidence appropriate to the conclusion, and a sufficient handoff. The errors here are synthetic; they were not found in Oleksiy's work. No novelty, independent review or measured improvement in AI reliability is claimed.

1. One observation, one decisive hypothesis

For $S \subseteq C_n = \mathbb{Z}/n\mathbb{Z}$, let

$$R_S(x) = \#\{(a,b) \in S^2 : a + b = x \pmod{n}\}.$$

Pairs are *ordered*, coefficients are exact integers, and the vector is labelled by residues. Recovery initially means equality of supports, not equality up to symmetries.

Correct statement. If n is odd, R_S determines S. Pair each off-diagonal (a,b) with (b,a). Their contribution at the same residue is even. Since doubling is bijective,

$$\boxed{\mathbf{1}_S(y) = R_S(2y) \pmod{2}.}$$

This is the general argument. Finite enumeration is corroboration within its checked range, not its proof.

Synthetic change. Remove the word "odd". In C_6, the different sets $S = \{0,1,2\}$ and $T = \{3,4,5\}$ satisfy

$$R_S = R_T = (1,2,3,2,1,0).$$

The unrestricted extension is false, but the original odd-order theorem survives. Moreover $T = S + 3$: this pair does *not* refute recovery up to translation.

2. The correction retains more than a warning

In C_{2m}, pairing still gives

$$R_S(2y) \equiv \mathbf{1}_S(y) + \mathbf{1}_S(y+m) \pmod{2}.$$

Thus the observation retains parity on antipodal pairs. If the support lies in a *known* set D containing at most one representative from each pair, exact recovery is restored. Merely knowing that at most one point of each pair is selected, without knowing the allowed representative, does not suffice.

Contribution to PAPP. This case records a true theorem, a false extension, a surviving identity, and a recovered subcase. It illustrates existing guidance rather than adding a mandatory record or claiming the guide lacks these distinctions.

CREDIT FOR THIS PAPP ADAPTATION

Worked example contributed by Alexandre Couret, prepared with AI assistance and adapted for PAPP by Oleksiy Babanskyy.

Companion edition 0.1.0. Adapted from the reviewed rc.2: page 5 states the attribution condition; page 6 separates companion commands from the historical LAB-03 record. Mathematical arguments and references are retained.

3. A complete criterion for the declared object class

Let G be any finite abelian group and $D \subseteq G$ a fixed, known allowed domain. The same cancellation proves

$$R_S(x) \equiv \sum_{2a=x} \mathbf{1}_S(a) \pmod 2.$$

Proposition. The map $S \mapsto R_S$ is injective on *all* subsets of D if and only if $a \mapsto 2a$ is injective on D.

Proof. If doubling is injective on D, the right-hand side at $x = 2a$, $a \in D$, contains at most one permitted nonzero term and recovers the indicator. Conversely, if $a \neq b$ in D satisfy $2a = 2b$, the singleton supports $\{a\}$ and $\{b\}$ have identical integer squares. □

Scope warning. Necessity uses singletons. It cannot be transferred unchanged to a class of supports of fixed size three, or to a previously discussed boundary theorem. Neither is audited here. The maximum such domain has $|2G| = |G|/|G[2]|$ elements, one from each nonempty fibre.

4. Prime-power extension, with explicit classical attribution

Let p be prime, $r \geq 1$, and $q = p^r$. For an integer function f on G, its q-fold convolution satisfies

$$\boxed{f^{*q}(x) \equiv \sum_{qa=x} f(a) \pmod p.}$$

In the commutative group algebra $\mathbb{F}_p[G]$, write $F = \sum_a \overline{f(a)} e_a$, where $e_a e_b = e_{a+b}$. The classical Frobenius identity and its iterates give

$$F^{p^r} = \sum_a \overline{f(a)}^{p^r} e_{p^r a} = \sum_a \overline{f(a)} e_{qa}.$$

The commutativity assumption and the coefficient field are used here. This is an elementary application of a classical identity, not a novelty claim [F1–F2].

Digit criterion. On the entire class of functions supported in known D with values in $\{0, \ldots, p-1\}$, $f \mapsto f^{*q}$ is injective exactly when $a \mapsto qa$ is injective on D. Sufficiency recovers each digit from the displayed formula. Necessity uses two colliding delta functions. If p is coprime to $|G|$, one can take $D = G$.

Decoder safeguard. For an arbitrary submitted data vector, residue decoding produces only a candidate. Recompute its *full integer* convolution power before accepting it. A matching residue vector alone is not an exact-data certificate.

Lost information. The reduced observation is a linear map over $\mathbb{F}_p$ of rank $|qG|$. Each attainable reduced vector has $p^{|G|-|qG|}$ preimages. These are *not* the fibres of the full integer observation, which retains more information.

5. Objections need exact scopes too

Translation versus reflection. In C_6, $A = \{0, 1, 4\}$ and $B = \{0, 2, 5\}$ share the square $(1, 2, 2, 0, 2, 2)$. They are not translates, but $B = -A$. They refute translation-only recovery, not recovery allowing reflections.

A stronger, finite witness. In C_8, set

$$A = \{0, 1, 3, 4\}, \qquad B = \{1, 2, 3, 6\}.$$

Then

$$R_A = R_B = (2, 2, 1, 2, 4, 2, 1, 2),$$

but no map $x \mapsto ux + t$, with u a unit modulo 8, maps A onto B.

A short human check complements the enumeration of all 32 affine maps. A has exactly one antipodal pair, $\{0, 4\}$, and B exactly one, $\{2, 6\}$. An affine map would force $t = 2$ or 6. However $\sum A \equiv 0$ and $\sum B \equiv 4 \pmod 8$, whereas the image sum would be $u \sum A + 4t \equiv 0$, a contradiction. No arbitrary graph-isomorphism conclusion follows.

Weights are not supports. In the odd-order group C_3, the different nonnegative weight vectors $(0, 1, 2)$ and $(2, 1, 0)$ both square to $(4, 4, 1)$. The indicator theorem is not an arbitrary-weight theorem.

Noncommutativity is substantive. In S_3, let $a = (12)$, $b = (23)$ and $f = \delta_a + \delta_b$. Then

$$f * f = 2\delta_e + \delta_{ab} + \delta_{ba}.$$

The cross terms are distinct and remain modulo 2. The naive diagonal identity fails; this is not a refutation of all possible nonabelian recovery methods.

Prime-power exponent is not prime-power modulus. In C_3, $(1, 1, 0)^{*4} = (5, 5, 6)$. Reducing modulo 4 does not give the pushforward along $a \mapsto 4a$. Iterated Frobenius uses exponent 4 and modulus 2.

6. Scope-aware checking

`check_witness.py` verifies equality of the integer squares and the equivalence requested by its caller. The same C_6 pair is accepted as an exact-recovery counterexample but rejected as a translation-recovery counterexample. This is deliberate.

`src/provenance.py` checks byte identity and reachability in a *declared* dependency graph only. A changed source may identify uses to inspect; it does not automatically make a theorem false. An unchanged hash does not make a theorem true. Missing formal evidence need not invalidate an elementary prose proof.

7. Executed checks and their limits

The replay uses Python integers and the standard library. The main results are deterministic; a repeated run is supplied in the preparation receipts.

Check	Coverage
Binary cyclic supports, all $2 \le n \le 14$	32,764
Known allowed domains, all $2 \le n \le 10$	2,044
Pairs (D, S) with $S \subseteq D$ in that domain check	88,569
Digit vectors: $p = 3$, $n \le 7$; $p = 5$, $n \le 5$	7,184
Binary vectors on four product groups at $p = 2, 3$	1,696
Iterated-prime-power binary checks	572
Local software unit tests	34
Selected deliberate faulty local behaviors detected	8

The product groups are $C_2 \times C_2$, $C_2 \times C_3$, $C_2 \times C_4$ and $C_3 \times C_3$. Their fixed-transversal checks are recorded separately. All numbers are local coverage, not independent statistical samples. The variants share a mathematical backbone and are not independent research domains.

What the eight fault controls mean. They exercise errors such as unordered counting, using differences instead of sums, dropping diagonals, using the wrong modulus, accepting residues without full-data recheck, and overextending a witness's scope. They are deliberately constructed tests of our code and checks. They are not eight mistakes found in PAPP, nor eight successful model trials.

What is not established. No independent mathematical review, no Lean compilation, no trial of the author's private agent, no FCIG audit and no comparative assistant or human handoff experiment. The elementary proofs are written and the finite computations are executed. Those are different kinds of evidence.

8. A test corpus without an invented model score

The kit provides 24 development prompts with separate organizer answers. They include positive controls, invalid extensions, insufficient evidence, unchanged semantics after renaming, and the separation of permissions from mathematical support.

The exporter creates three local packets: task only, task plus a short conventional checklist, and task plus selected frozen PAPP guidance. Answer keys, explanatory reports, results and draft correspondence are excluded by an explicit allow-list. This separation is tested mechanically. It prepares evaluation; it does not prove blinding, nor start an assistant.

`grade.py` compares categorical labels only. Explanations, proofs and the scope of retained results require human adjudication. An all-refuse policy must be penalized for rejecting valid claims; low false-acceptance alone is not success.

9. The next useful experiment is reopening

Give a consenting new reader only the minimal handoff and named artifacts, not the original conversation. Before the trial, fix the retrieval questions: the exact surviving claim, the rejected extension, the proof location, actual finite coverage, missing formal evidence and next authorized action. Record help requested and time as well as correctness.

The included before/after handoff is a prepared exercise, **not an executed fresh-context trial**. A different Python process does not count as a new reader. An assistant that authored the cases cannot become unaware of them by changing its role label.

10. A fair comparison, if separately agreed

Keep the model version, tools, problem inputs, permissions and resource budgets comparable across the three conditions. Counterbalance order and retain exact outputs, costs and failures. Grade explanations without knowing the condition. Use the current known cases to debug the experiment, then select genuinely new, independently curated problems for a later confirmatory test.

Report false acceptance, false rejection, preservation of unaffected results, handoff performance and verification burden separately. Repeated variants in one family are clustered observations, not an enlarged independent sample. Fix sample size and stopping rules before a real study; no significance or power claim is made here.

If a simple checklist performs as well as PAPP, record that. If PAPP helps reopening but not mathematical correctness, state that narrower contribution. If it adds burden or errors, preserve that result too. This proposal assumes no favorable outcome.

11. A small addition Oleksiy could choose

The guide already says to preserve surviving lemmas and reconcile affected uses [P1–P3]. A modest integration could be one optional worked-example page linked from the mathematics module. It could use the C_6 example first, the known-domain criterion second, and the scope-aware C_8 witness only as an extension.

No new registry, PAPP rewrite, collaboration agreement, coauthorship or review campaign is required. Public reuse must retain Alexandre's approved credit and mathematical references. His later permission supersedes the original attribution-waiver wording. The evaluation infrastructure remains optional.

Delivery boundary. This is a local candidate contribution prepared for Alexandre. No email, pull request, comment, review, public release or external workspace modification has been made. No private correspondence or FCI material is included. The upstream selection retains Oleksiy's MIT license; a public release of our new contribution remains a separate decision.

12. Commands for this companion

From the companion root:

```
python check_examples.py
```

From supplementary/code/:

```
python replay.py --output my-run.json
```

Use Python 3.10+ and choose a new output path. These commands are available in this 12-file companion; they call no external service.

Historical LAB-03 record (not companion instructions)

The original run_all.py, check_witness.py, verify_manifest.py and unit-test workflow are not distributed here. The source record below describes that original kit, not this companion.

Source version. The selected public PAPP documents were read at commit

`ed4a392af8f09ec8fc7b5b3bdd2452bbd246722b`.

The repository's main branch was checked to point to that commit during preparation. The delivery identifies itself as PAPP/github-1.0.3 and PAPP/documentary-1.0.3 [P3]. The selected UTF-8 texts were reconstructed from connector output and verified against their Git blob hashes; SHA256 and sizes are recorded in `upstream/SOURCE_SNAPSHOT.json`. This is byte-identity checking, not author authentication or proof of scientific truth.

References

[P1] Oleksiy Babanskyy, *PAPP — the small working core*, pinned version above. Exact source.

[P2] Oleksiy Babanskyy, *Mathematics: preserve what the result actually says*, same commit. Exact source.

[P3] Oleksiy Babanskyy, *What this edition offers*, *Documentary working agreement*, and *Use the contract the project actually has*, same commit. Selected exact texts and original license are included under `upstream/`. They expressly distinguish documentary guidance from measured general efficacy or automatic semantic validation.

[F1] The Stacks Project, Section 33.36, *Frobenii*, tag 0CC6. Classical Frobenius background; not a novelty search for this worked example.

[F2] Mathlib documentation, `Mathlib.Algebra.CharP.Frobenius`. Reference for Frobenius and its iterates. No Lean implementation of the present proposition was compiled.

Review request, not validation request

A useful next response from Oleksiy would concern whether this worked example fits his intended use of PAPP. A more extensive comparison should be separately agreed. Neither a favorable reply nor an accepted example would certify the wider Couret–Unification program.